

Q-Transplant: AI-Driven Quantum-Inspired Organ Matching and Smart Transplant Coordination System

ABSTRACT

Organ transplantation is a life-saving medical procedure that requires accurate donor-recipient matching, rapid transportation, and efficient coordination among healthcare institutions. Traditional transplant management systems often face challenges such as delayed communication, inefficient organ allocation, limited monitoring, and lack of real-time coordination. To address these issues, Q-Transplant is proposed as an intelligent organ transplant coordination platform that integrates Artificial Intelligence (AI), Internet of Things (IoT), and Quantum-Inspired optimization techniques.The system uses a Quantum-Inspired Matching Engine to evaluate donor-recipient compatibility based on factors such as blood group, HLA compatibility, medical urgency, geographical distance, organ ischemia time, and hospital resource availability. It generates an optimized compatibility score to support informed clinical decision-making. The platform provides secure portals for administrators, hospitals, doctors, donors, and patients, enabling seamless registration, verification, organ allocation, and emergency coordination.Q-Transplant incorporates ESP32-based IoT devices to monitor organ transportation in real time by tracking GPS location, temperature, battery status, and emergency alerts. Additional features include AI-assisted clinical decision support, ambulance and hospital tracking, automated notifications, role-based authentication, interactive dashboards, and real-time communication. Developed using FastAPI, SQLAlchemy, JavaScript, HTML, CSS, and PostgreSQL, the system enhances transparency, reduces response time, and improves the efficiency and success rate of organ transplantation processes.

Keywords: Artificial Intelligence (AI), Organ Transplantation, Quantum-Inspired Matching, Internet of Things (IoT), Clinical Decision Support.

Team Members **Project Guide**

Aravindh A Ms.Hariprabha

Jovin Roy S

Priyan E